

Отчет по лабораторной 3 «Yolo»

```
1 # Path to the dataset root directory
2 train: train
3 test: test
4 val: val
5 nc: 3
6 names:
7   0: dog
8   1: person
9   2: cat
```

Рисунок 1 — Файл конфигурации

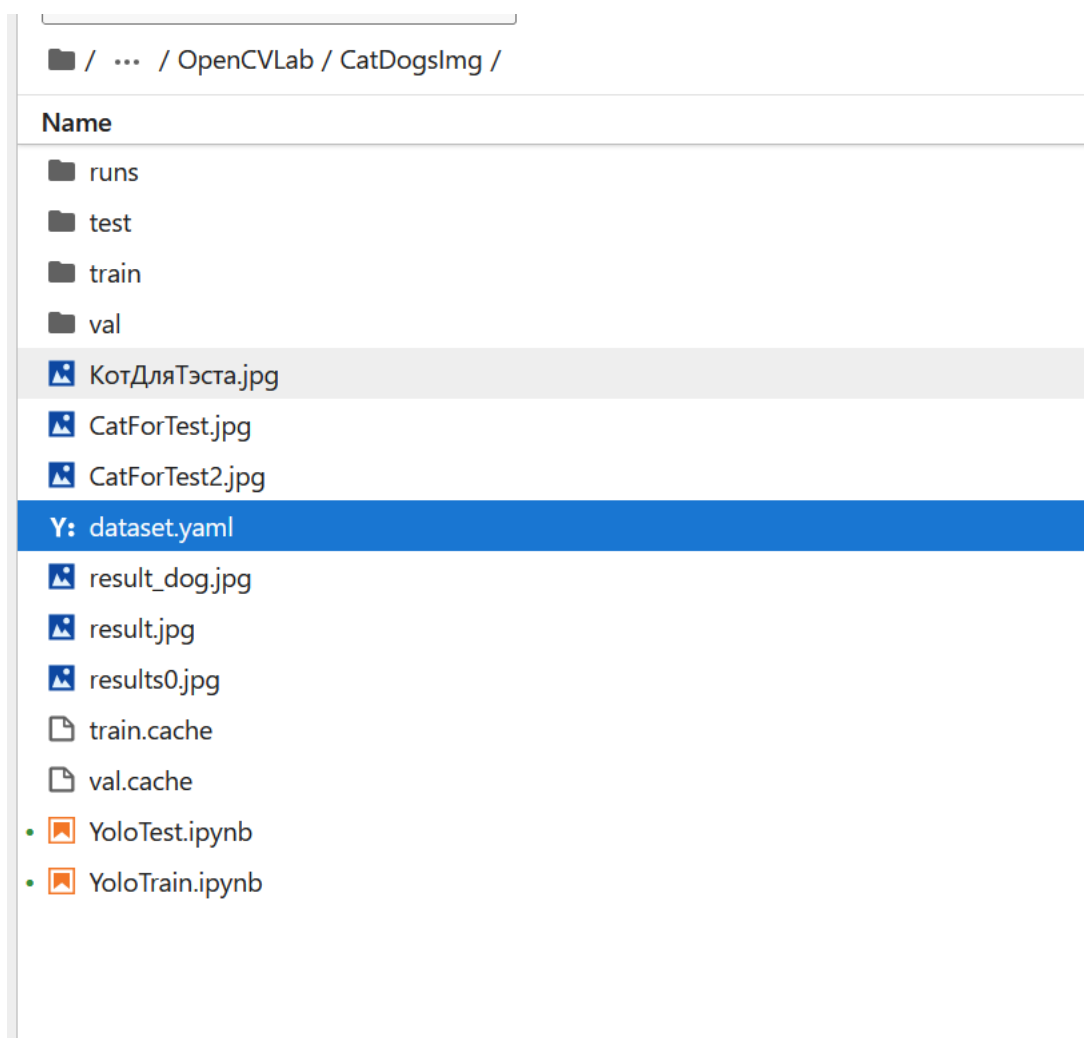


Рисунок 2 — Структура проекта

Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
28/30	0G	1.843	3.308	0.03545	10	640: 100%	18/18 7.8s/it 2:203.8ss
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	16	45	0.0111	0.979	0.142	1/1 2.3s/it 2.3s
						0.0491	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
29/30	0G	1.856	3.276	0.03442	10	640: 100%	18/18 8.4s/it 2:314.2ss
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	16	45	0.00962	0.979	0.124	1/1 3.9s/it 3.9s
						0.0443	
Epoch	GPU_mem	box_loss	cls_loss	dfl_loss	Instances	Size	
30/30	0G	1.873	3.224	0.03599	10	640: 100%	18/18 8.2s/it 2:273.8ss
	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
	all	16	45	0.00989	1	0.131	1/1 2.6s/it 2.6s
						0.0457	

0 epochs completed in 1.286 hours.
 optimizer stripped from /home/jovyan/OpenCV_Projects/OpenCVLab/CatDogsImg/runs/detect/train-6/weights/last.pt, 5.4MB
 optimizer stripped from /home/jovyan/OpenCV_Projects/OpenCVLab/CatDogsImg/runs/detect/train-6/weights/best.pt, 5.4MB

validating /home/jovyan/OpenCV_Projects/OpenCVLab/CatDogsImg/runs/detect/train-6/weights/best.pt...
 ultralytics 8.4.45 Python-3.12.7 torch-2.5.1+cpu CPU (11th Gen Intel Core i5-1135G7 @ 2.40GHz)
 YOLO26n summary (fused): 122 layers, 2,375,421 parameters, 0 gradients, 5.2 GFLOPs

Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	
all	16	45	0.00886	0.917	0.158	0.0709	1/1 2.0s/it 2.0s
dog	7	21	0.00729	1	0.261	0.121	
cat	7	24	0.0104	0.833	0.0545	0.0208	

speed: 2.1ms preprocess, 118.1ms inference, 0.0ms loss, 0.1ms postprocess per image
 results saved to /home/jovyan/OpenCV_Projects/OpenCVLab/CatDogsImg/runs/detect/train-6
 ultralytics.utils.metrics.DetMetrics object with attributes:

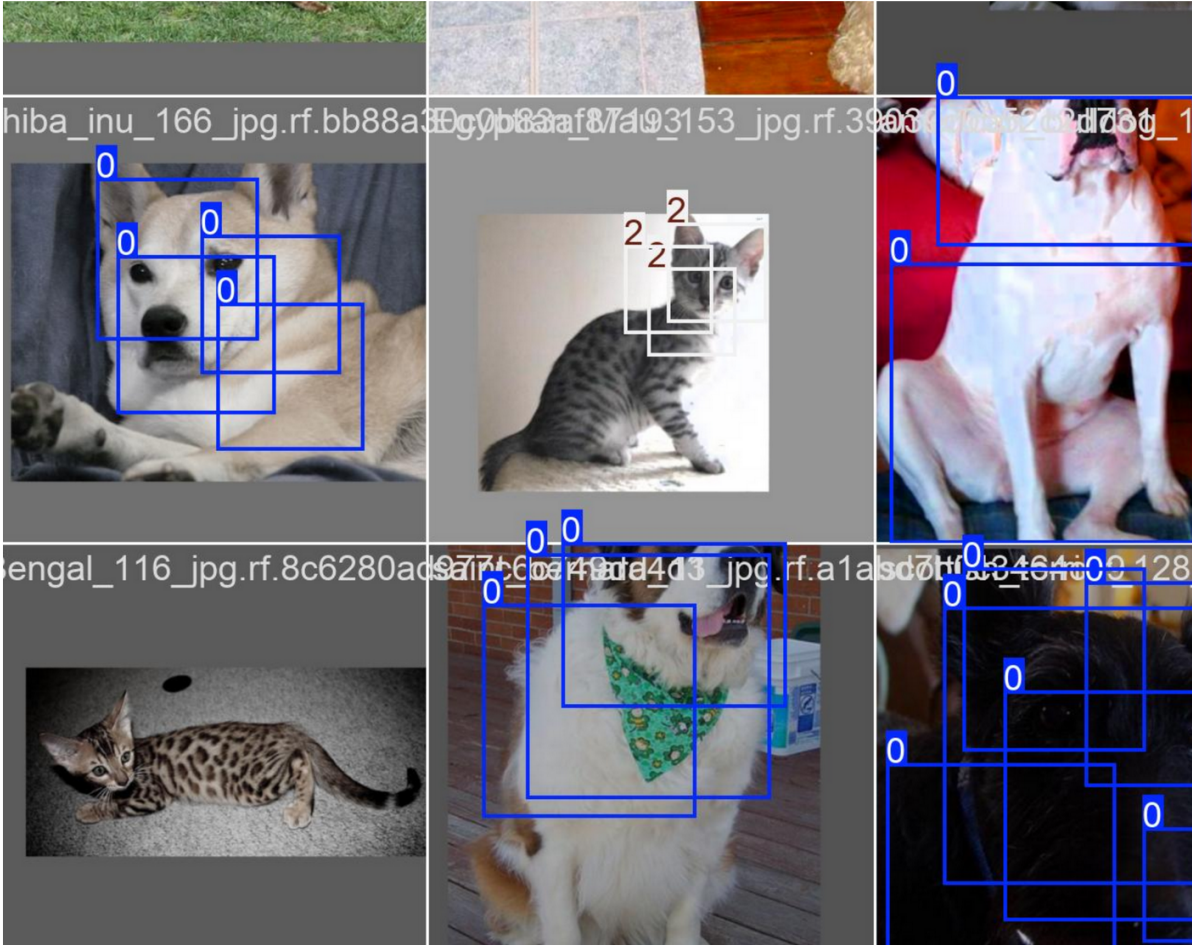
p_class_index: array([0, 2])
 m: ultralytics.utils.metrics.Metric object
 confusion_matrix: <ultralytics.utils.metrics.ConfusionMatrix object at 0x7a9560199a60>
 curves: ['Precision-Recall(B)', 'F1-Confidence(B)', 'Precision-Confidence(B)', 'Recall-Confidence(B)']
 curves_results: [[array([
 0, 0.001001, 0.002002, 0.003003, 0.004004, 0.005005, 0.006006, 0.007007,
 0.008008, 0.009009, 0.01001, 0.011011, 0.012012, 0.013013, 0.014014, 0.015015, 0.016016, 0.017017, 0.0
 018, 0.019019, 0.02002, 0.021021, 0.022022, 0.023023,
 0.024024, 0.025025, 0.026026, 0.027027, 0.028028, 0.029029, 0.03003, 0.031031, 0.032032, 0.033
 3, 0.034034, 0.035035, 0.036036, 0.037037, 0.038038, 0.039039, 0.04004, 0.041041, 0.042042, 0.043043
 .044044, 0.045045, 0.046046, 0.047047,
 0.048048 0.049049 0.05005 0.051051 0.052052 0.053053 0.054054 0.055055 0.056056 0.057
]

Рисунок 4 — результаты обучения

085,	0.086086,	0.087087,	0.088088,	0.089089,	0.09009,	0.091091,	0.092092,	0.093093,	0.094094,	0.095095,		
911,	0.096096,	0.097097,	0.098098,	0.099099,	0.1001,	0.1011,	0.1021,	0.1031,	0.1041,	0.10511,	0.10611,	0.10711,
	0.11011,	0.11111,	0.11211,	0.11311,	0.11411,	0.11512,	0.11612,	0.11712,	0.11812,	0.11912,		
	0.12012,	0.12112,	0.12212,	0.12312,	0.12412,	0.12513,	0.12613,	0.12713,	0.12813,	0.12913,	0.13013,	0.13113,

Результаты обучения(30 эпох):

![[изображение.png](attachment:ddf8290b-281e-4b1a-8ec1-320e8af8b27c.png)]]



Метрики

Рисунок 5- результаты обучения

	epoch	time	train/box_loss	train/cls_loss	train/dfl_loss	metrics/precision(B)	metrics/recall(B)	metrics/mAP50(B)	metrics/mAP50-95(B)	
1	1	163.402	3.15746	5.52733	0.04926	0.00125	0.28571	0.00074	0.00016	
2	2	325.017	3.30354	5.37046	0.05356	0.00078	0.125	0.00026	5e-05	
3	3	480.996	3.28604	5.18357	0.05315	0.00084	0.16667	0.00028	4e-05	
4	4	636.159	3.21257	5.06108	0.05337	0.00156	0.35714	0.00132	0.00052	
5	5	795.386	2.98984	4.62571	0.04662	0.00146	0.33333	0.00105	0.00025	
6	6	949.766	2.85786	4.48148	0.04499	0.00052	0.11905	0.00013	4e-05	
7	7	1101.82	2.82547	4.19436	0.04426	0	0	0	0	
8	8	1262.81	2.83518	4.13317	0.047	0.00025	0.02083	4e-05	1e-05	
9	9	1417.65	2.75518	3.98835	0.04364	0.00664	0.60714	0.00485	0.00126	
10	10	1568.33	2.70017	3.79439	0.04275	0.00715	0.6994	0.01797	0.00466	
11	11	1714.64	2.59066	3.634	0.04069	0.00516	0.45536	0.00649	0.00141	
12	12	1879.96	2.55775	3.59978	0.03913	0.00513	0.45536	0.00909	0.00236	
13	13	2050.28	2.50757	3.62632	0.03787	0.01041	0.57143	0.03654	0.00865	
14	14	2214.43	2.38493	3.47231	0.03647	0.00737	0.71726	0.0159	0.00496	
15	15	2375.03	2.28899	3.38847	0.03563	0.00974	0.2619	0.02836	0.0083	
16	16	2532.99	2.31235	3.39613	0.03597	0.53386	0.19048	0.04771	0.01168	
17	17	2682.88	2.27896	3.27847	0.03479	0.52457	0.2381	0.0422	0.01634	
18	18	2835.6	2.28837	3.19565	0.03438	0.59139	0.07143	0.05695	0.02094	
19	19	2983.1	2.27202	3.24473	0.03571	0.5315	0.02381	0.06257	0.03231	
20	20	3137.34	2.20253	3.19301	0.03368	0.01705	0.2381	0.04113	0.01432	
21	21	3283.83	2.09734	3.68716	0.04019	0.54593	0.14286	0.09552	0.04706	
22	22	3428.26	1.93408	3.61135	0.03732	0.00834	0.84524	0.11392	0.04907	
23	23	3578.41	1.94271	3.46066	0.03686	0.00886	0.91667	0.15776	0.07083	
24	24	3726.19	1.97063	3.43105	0.03784	0.00905	0.9375	0.14604	0.06623	
25	25	3870.71	1.91568	3.35849	0.03719	0.01234	0.97619	0.12969	0.06889	
26	26	4022.46	1.94727	3.29181	0.03838	0.03622	0.82143	0.1401	0.06225	
27	27	4180.15	1.88983	3.32957	0.03556	0.02106	0.88988	0.14291	0.0521	
28	28	4322.61	1.84331	3.30805	0.03545	0.01112	0.97917	0.14181	0.04908	
29	29	4477.86	1.85627	3.27604	0.03442	0.00962	0.97917	0.12419	0.04434	
30	30	4628.17	1.87261	3.22419	0.03599	0.00989	1	0.13091	0.0457	

Рисунок 6 — таблица метрик обучения

Графики

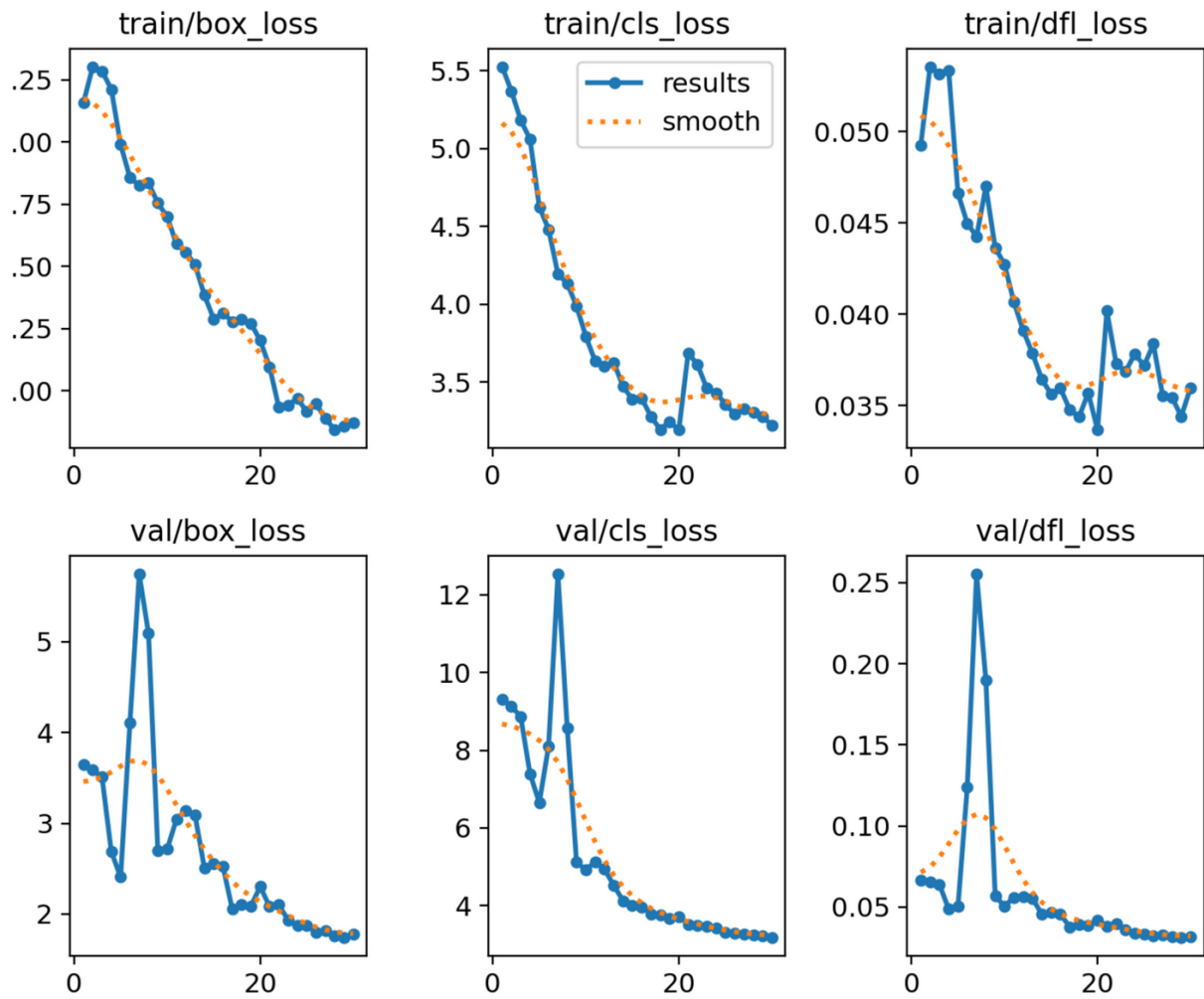


Рисунок 7 — графики обучения

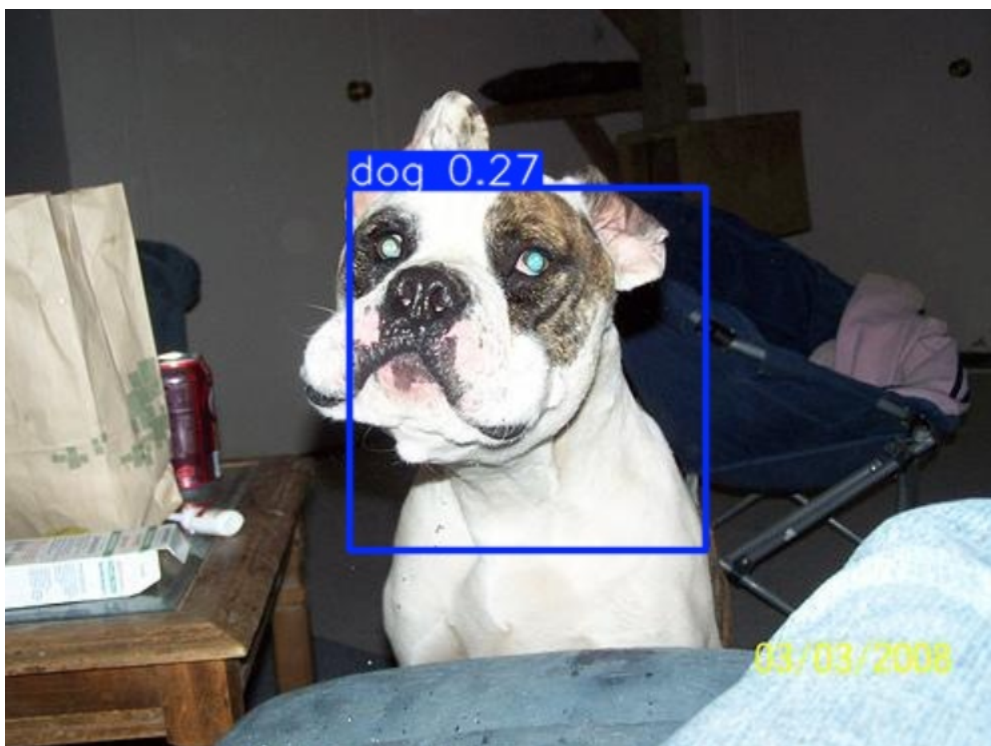


Рисунок 8- Результат распознавания

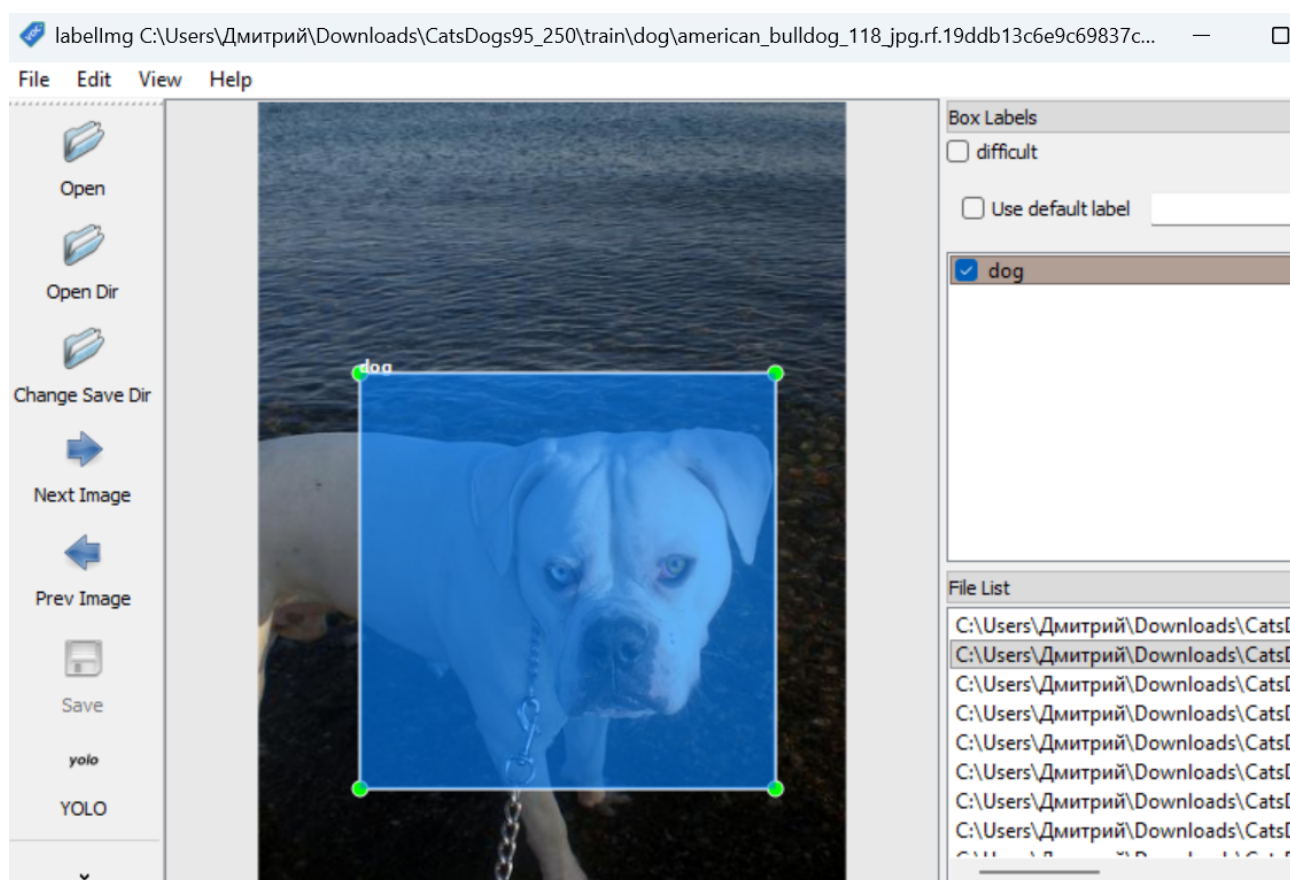


Рисунок 9 — пример разметки изображения с помощью labelimg

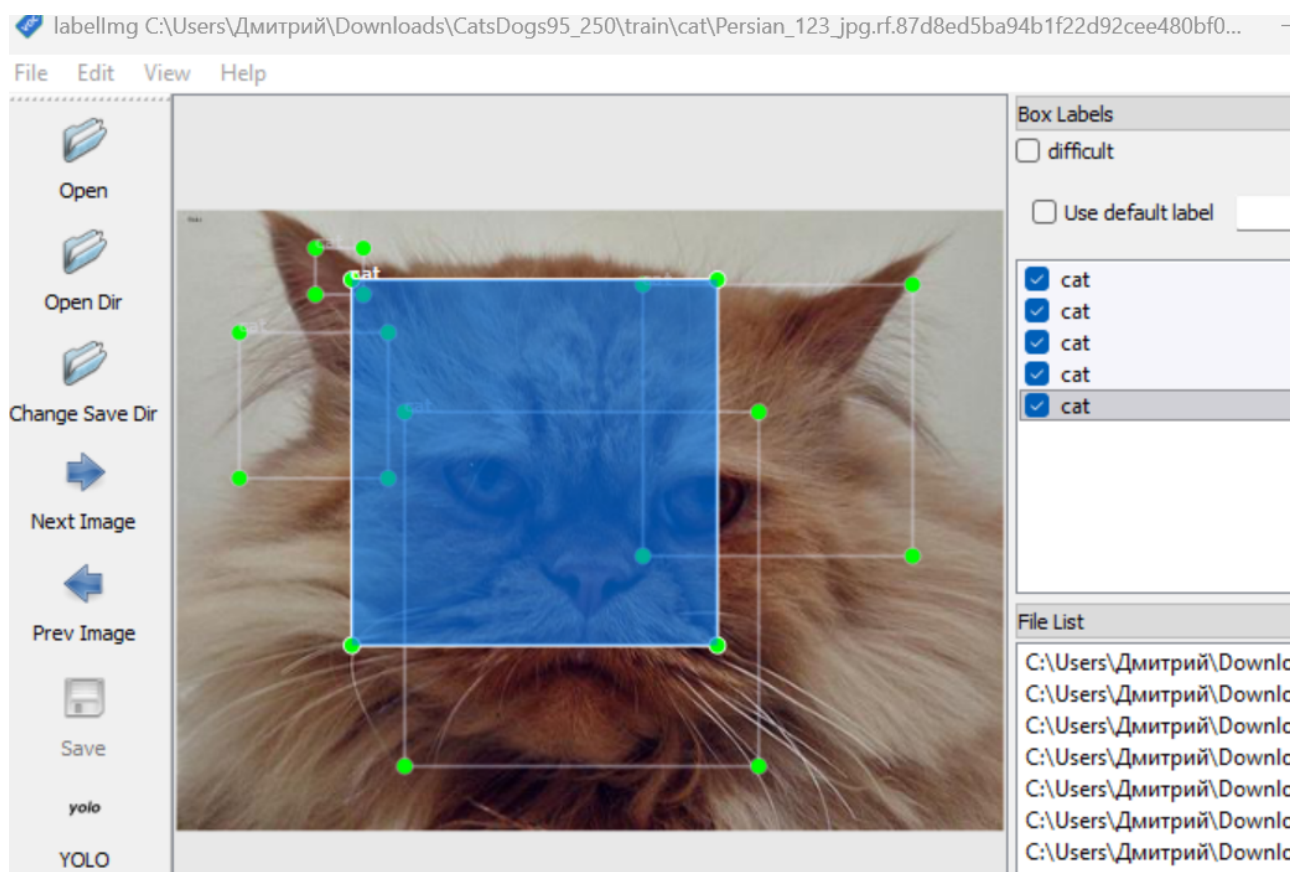


Рисунок 9 — пример разметки изображения с помощью labeling